

MATRIX CHAIN OPTIMIZER 3D

Interval Dynamic Programming Visualizer

Outputs:

INTERVAL DYNAMIC PROGRAMMING

MATRIX CHAIN OPTIMIZER

Visualize optimal matrix multiplication using dynamic programming.

01

MATRIX CONFIGURATION

Choose the number of matrices and enter their dimensions.

NUMBER OF MATRICES

3

BUILD MATRICES

DIMENSIONS

N MATRICES = N + 1 DIMENSIONS

ROWS

5

D1

10

D2

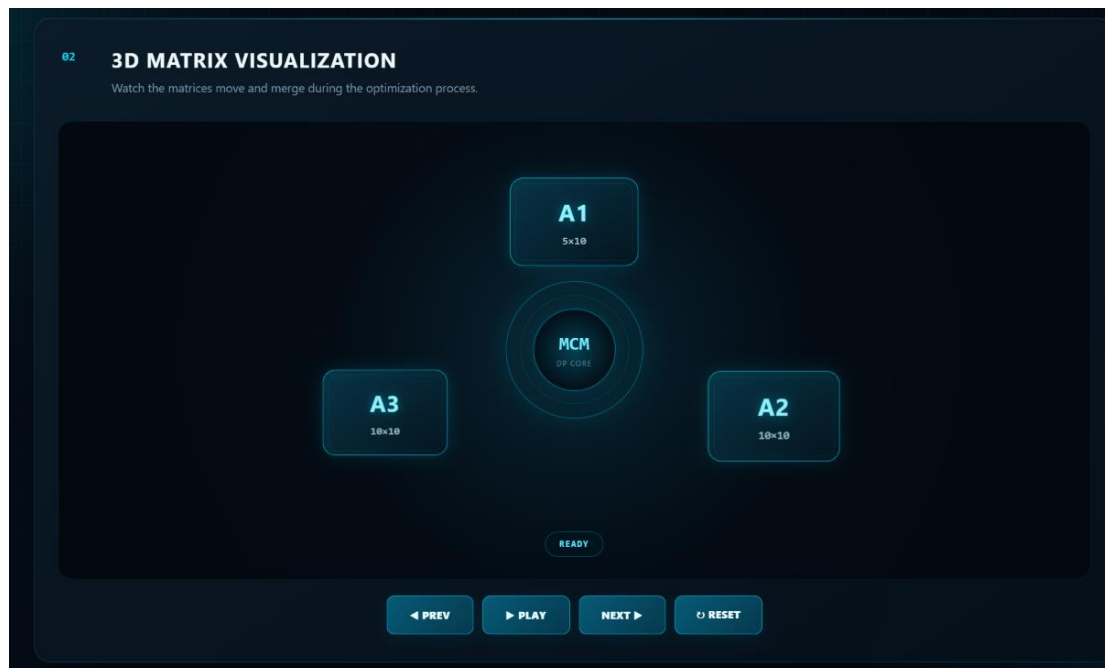
10

FINAL COLS

10

MATRIX CHAIN

A1 × A2 × A3



02

3D MATRIX VISUALIZATION

Watch the matrices move and merge during the optimization process.



STEP 2 / 2

◀ PREV

▶ PLAY

NEXT ▶

↺ RESET

03

OPTIMIZATION

Calculate the minimum scalar multiplication cost.

OPTIMIZE MATRIX CHAIN

04

OPTIMIZATION RESULTS

Comparison between optimized and naive multiplication.

OPTIMIZED COST

1,000

SCALAR MULTIPLICATIONS

NAIVE COST

1,000

LEFT → RIGHT

OPERATIONS SAVED

0

OPERATIONS

EFFICIENCY

0.00%

IMPROVEMENT

OPTIMAL PARENTHESIZATION

 $((A1 \times A2) \times A3)$

05

DP COST TABLE

Minimum multiplication cost for every matrix interval.

0	500	1,000
-	0	1,000
-	-	0

DP FORMULA $dp[i][j] = \min(dp[i][k] + dp[k+1][j] + p[i-1] \times p[k] \times p[j])$